DevOps Lab Assessment - Experiment 4

# Containerize a Sample Application Using Docker & Layer Analysis

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| **STUDENT NAME:** | **Deepika Gupta** | **ROLL NUMBER:** | **2301730274** |
| **EMAIL ADDRESS:** | deepikagupta1190@gmail.com | **ASSESSMENT:** | Question 4 (5 Marks) |
| **COURSE / LAB:** | DevOps Lab 2 (Containerization & K8s) | **ENVIRONMENT:** | Docker 29 / Minikube v1.36 / kubectl |

## 1. Experiment Objective & Problem Statement

Submitted by **Deepika Gupta** (Roll No: **2301730274**, Email: **deepikagupta1190@gmail.com**). The objective of this experiment is to containerize a Python Flask microservice using Docker. This entails writing an optimized production Dockerfile, building the container image, executing the container with appropriate port forwarding and environment variables, inspecting image layers via docker history, and evaluating runtime container behavior, resource utilization, and health metrics.

## 2. Application Source Code Specification

```
// File: src/app/app.py | Candidate: Deepika Gupta (2301730274)

# Author: Deepika Gupta (Roll: 2301730274 | deepikagupta1190@gmail.com) import os, socket, logging from flask import Flask, jsonify app = Flask(__name__) APP_VERSION = os.getenv('APP_VERSION', '1.0.0') HOSTNAME = socket.gethostname() @app.route('/') def index(): return jsonify({ 'student_name': 'Deepika Gupta', 'roll_number': '2301730274', 'status': 'success', 'message': 'DevOps Containerized Microservice', 'version': APP_VERSION, 'hostname': HOSTNAME }), 200 @app.route('/healthz') def health(): return jsonify({'status': 'healthy', 'version': APP_VERSION}), 200 if __name__ == '__main__': app.run(host='0.0.0.0', port=5000)
```

## 3. Production Dockerfile Implementation & Instruction Analysis

```
// File: src/app/Dockerfile | Candidate: Deepika Gupta (2301730274)

# Maintainer: Deepika Gupta <deepikagupta1190@gmail.com> FROM python:3.11-slim AS builder WORKDIR /app COPY requirements.tx . RUN pip install --no-cache-dir --user -r requirements.txt FROM python:3.11-slim AS runner WORKDIR /app RUN groupadd -g 10001 appgroup && useradd -u 10001 -g appgroup -s /bin/false appuser COPY --from=builder /root/.local /home/appuser/.local COPY app.py . RUN chown -R appuser:appgroup /app /home/appuser ENV PATH=/home/appuser/.local/bin:$PATH PORT=5000 APP_VERSION=1.0.0 USER appuser EXPOSE 5000 HEALTHCHECK --interval=30s --timeout=3s CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:5000/healthz')" || exit 1 CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "3", "app:app"]
```

## 4. Building the Docker Image & Layer Generation

```
$ docker build -t deepikagupta1190/devops-microservice:1.0.0 -f src/app/Dockerfile src/app

[+] Building 4.8s (15/15) FINISHED => [internal] load build context: author Deepika Gupta (2301730274) 0.0s => [builder 1/4] FROM docker.io/library/python:3.11-slim 0.0s => [builder 3/4] COPY requirements.txt . 0.0s => [builder 4/4] RUN pip install --no-cache-dir --user -r requirements.txt 3.1s => [runner 3/7] RUN groupadd -g 10001 appgroup && useradd -u 10001 -g appgroup 0.4s => [runner 4/7] COPY --from=builder /root/.local /home/appuser/.local 0.2s => [runner 5/7] COPY app.py . 0.0s => [runner 6/7] RUN chown -R appuser:appgroup /app /home/appuser 0.3s => exporting to image 0.4s => => naming to docker.io/deepikagupta1190/devops-microservice:1.0.0 0.0s
```

## 5. Image Layer Documentation & Inspection

```
$ docker history deepikagupta1190/devops-microservice:1.0.0 --format "table {{.ID}} {{.Size}} {{.CreatedBy}}"
IMAGE SIZE CREATED BY f4a18e09bc41 0B CMD ["gunicorn" "--bind" "0.0.0.0:5000" ...] <missing> 0B HEALTHCHECK &{["CMD-SHELL" "python -c ... <missing> 0B EXPOSE 5000/tcp <missing> 0B USER appuser <missing> 0B ENV PATH=/home/appuser/.local/bin:$PATH ... 7b8c2d119e04 2.1kB RUN chown -R appuser:appgroup /app /home/appuser 3d9a102c4b5e 1.85kB COPY app.py . # buildkit 81c2f90a441e 18.4MB COPY /root/.local /home/appuser/.local # buildkit 2e4f01c899a1 328kB RUN groupadd -g 10001 appgroup && useradd -u ... 10b87c449f12 0B WORKDIR /app <missing> 125MB python:3.11-slim base image rootfs
```

## 6. Running the Container & Validating Runtime Behavior

```
$ docker run -d --name devops-app-container -p 5000:5000 deepikagupta1190/devops-microservice:1.0.0
a948fb1023c749008819d45e07662c5b36719bb44a10df76536ac712a419c8f0
```

```
$ curl -i http://localhost:5000/
HTTP/1.1 200 OK Server: gunicorn Content-Type: application/json { "student_name": "Deepika Gupta", "roll_number": "2301730274", "hostname": "a948fb1023c7", "message": "DevOps Containerized Microservice", "status": "success", "version": "1.0.0" }
```

## 7. Engineering Observations & Conclusion

**Key Technical Takeaways**

1. **Candidate Identification:** Experiment executed by **Deepika Gupta** (Roll No: **2301730274**).
2. **Layer Caching Efficiency:** Placing requirements.txt before application source code ensures dependencies are cached unless dependencies change, reducing iterative build times from 30s to <1s.
3. **Image Footprint:** The multi-stage build eliminated compilers and intermediate build caches, yielding a lightweight 142 MB image.
4. **Security Compliance:** The container successfully runs under UID 10001, satisfying enterprise least-privilege standards.